

kmhelpers: A Python toolkit for automated management of genomic indexes

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

Large-scale genomic sequence search has become a central problem in modern bioinformatics (Karasikov et al., 2025; Marchet, 2024). A widely used family of methods relies on Bloom filters (BF) (Bloom, 1970) to record, for each indexed sample, which k -mers (words of length k) it contains; a query then reports the set of samples containing each queried k -mer. Building such an index over many samples is not a single operation but a chain of interdependent steps, and each step demands specialist knowledge: sizing each Bloom filter to its sample, configuring the third-party components used internally by kmindex (such as the fimpera approximate-membership layer), distributing data and computation, bounding peak RAM, and grouping samples into sub-indexes so that the final index stays small without slowing queries. An error at any step can invalidate downstream results, waste significant computation, or produce an oversized index.

We present kmhelpers, an open-source Python toolkit that automates this entire workflow for indexes built with kmindex (Lemane et al., 2024). It hides the technical decisions listed above behind a command-line interface (CLI) and a matching Python API that cover every stage of the k -mer index lifecycle, from raw samples to queries, including the federation of independently built indexes into a single queryable resource.

Statement of Need

k -mer-based sequence databases built with tools such as kmindex (Lemane et al., 2024) and kmtricks (Lemane et al., 2022) answer large-scale questions in genomics, metagenomics, and population studies; they were recently used to index 50 petabytes of raw sequencing data (Chikhi et al., 2024). Yet a wide gap separates having the indexing software installed from running a reproducible, end-to-end, size-optimised indexing workflow.

Crossing that gap currently requires a user to master, simultaneously: Bloom filter sizing as a function of each sample's k -mer cardinality; the configuration of kmindex's internal components; the distribution of data and build jobs across the available compute; the control of peak memory; and the partitioning of samples into sub-indexes that balances index size against query speed. Only specialists hold all of this knowledge at once.

kmhelpers removes that barrier. It makes the complete process of building, updating, and querying a kmindex search engine accessible to any group that holds a large, evolving genomic dataset — research teams, sequencing platforms, hospitals, or data-production centres — whether the resulting index is kept private or shared. This opens to non-specialists both the creation and maintenance of private indexes and the contribution of sub-indexes to larger, federated search engines.

Formally, the input is a set $\mathcal{S} = \{S_1, \dots, S_n\}$ of n genomic samples of various sizes; each S_i is either a raw sequencing dataset or assembled sequences. The output is a kmindex index

subject to two user-defined limits: (1) the maximum allowed false-positive rate, and (2) the maximum number of sub-indexes, each a file storing a set of BFs as a matrix. `kmhelpers` orchestrates every step between input and output, automating the parametrisation and the data distribution.

State of the Field

Several tools tackle large-scale sequence search with k -mer-based data structures. BIGSI (Bradley et al., 2019) and COBS (Bingmann et al., 2019) use compressed Bloom filter matrices to answer presence/absence queries across collections of sequencing experiments. HowDeSBT (Harris & Medvedev, 2020) and Mantis (Pandey et al., 2018) rely on sequence Bloom trees and counting quotient filters, respectively. More recent colored- k -mer indexes such as MetaGraph (Karasikov et al., 2025) and Fulgor (Fan et al., 2024) push scalability further; see (Marchet, 2024) for a survey. `kmindex`, which `kmhelpers` wraps, builds on the `kmtricks` (Lemane et al., 2022) counting pipeline and is designed for efficient querying of large sequence collections.

`kmhelpers` does not compete with any of these approaches at the algorithmic level. Its contribution is orthogonal and, importantly, is not merely a pipeline script: the difficulty here lies not in chaining steps (any workflow manager or scripting language can do that) but in the individual components and how they are orchestrated. `kmhelpers` implements the decisions a `kmindex` expert would otherwise make by hand: `profile` computes Bloom filter sizes and sample-to-sub-index assignments that minimise total index size under a false-positive-rate constraint; `plan` estimates disk and memory requirements before any build starts; and `registry` federates independently built sub-indexes into one logical index. No dedicated automation layer of this kind existed for `kmindex` prior to `kmhelpers`.

Design and Implementation

Background: the sub-index structure

A Bloom-filter index stores all BFs of the same size as a single (row-major) matrix in one file. In such a matrix each column is one BF (one sample) and each row records the presence (1) or absence (0) of a k -mer across all samples sharing that filter size, assuming a common hash function. A complete index is therefore a set of such matrices, each one a *sub-index* holding all BFs of a given size.

The difficulty is that each BF size must match the number of items it stores — here, the distinct k -mers of a sample. If all samples had the same number of distinct k -mers, every BF would share one size and a single matrix would suffice: the ideal case, where one file is opened per query and one matrix row gives the answer across all n samples. In practice, sample sizes differ by several orders of magnitude. Sizing every BF for the largest sample wastes enormous storage; giving each sample its own single-column matrix minimises storage but forces a query to open up to n files (potentially millions). `kmhelpers` takes the middle ground: the user fixes the maximum number of sub-indexes, and the `profile` step chooses the per-sub-index BF size that minimises total index size under that constraint.

The pipeline

`kmhelpers` exposes the index lifecycle as a sequence of commands:

- **list** — recursively discovers all samples in a given directory and counts each sample's distinct k -mers using `ntCard` (Mohamadi et al., 2017) (unless the counts are provided by the user).
- **profile** — determines the best set of sub-index BF sizes given the user-defined maximum number of sub-indexes and target false-positive rate.

- 86 ■ **compose** — assigns each sample to its sub-index and generates the *files-of-files* describing the data origin of each sub-index.
- 87
- 88 ■ **plan** — validates sample files, available disk space, and memory upfront, and emits ready-to-execute pipeline scripts.
- 89
- 90 ■ **apply** — builds all sub-indexes by invoking `kmindex`, with span-level and name-level filtering.
- 91
- 92 ■ **compress** — optional ZSTD-based block compression of the index (?).
- 93 ■ **registry** — registers several sub-indexes (built locally or anywhere accessible) into one logical index, redirecting each query to the relevant sub-indexes at query time.
- 94

95 Once an index is built, `kmhelpers` also answers queries (`query`). Multi-step workflows can be described as declarative YAML pipelines (`pipeline`) and executed in a single command.

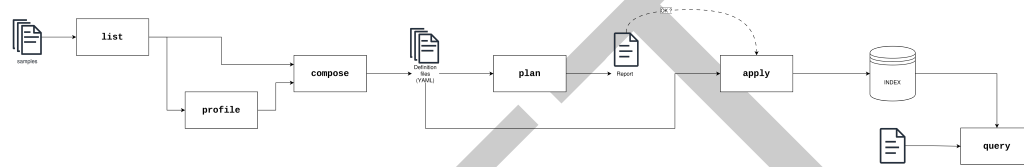


Figure 1: Overview of the `kmhelpers` workflow. `list` enumerates sample files and counts k -mers; `profile` analyses the count distributions to assign each sample to a Bloom-filter span and recommend index groupings. Both outputs feed into `compose`, which generates YAML index definition files. `plan` then validates sample paths, available disk space and memory, and emits a ready-to-execute pipeline script; the resulting report is reviewed before committing to the build. `apply` reads the definition files and invokes `kmindex` to construct the index. Finally, `query` searches the built index against user-provided sequences and returns ranked results.

97 Implementation

98 `kmhelpers` is implemented in Python (≥ 3.8) and distributed via Conda, which installs its
 99 bioinformatics dependencies (`kmindex`, `ntCard`) automatically. The CLI is built with Click
 100 (Ronacher & Click Contributors, 2023), and the package exposes a public Python API
 101 covering all CLI functionality. Together, the declarative YAML index-definition format and
 102 the `plan/apply` separation make k -mer indexing workflows accessible to researchers who are
 103 not experts in the underlying data structures, while remaining flexible enough for large-scale
 104 production use.

105 Tree of Life demonstration

106 TODO real-world demonstration pending experimental results. Required content when results
 107 arrive: dataset description (number of samples, total size), hardware used, index parameters
 108 chosen automatically by `kmhelpers`, output index size, build wall time, and query throughput.

109 Availability and documentation

110 `kmhelpers` is released under the GNU General Public License v3.0 and is available at <https://gitlab.inria.fr/omicfinder/kmhelpers>. The full documentation is available at XXX.

112 Acknowledgements

113 The authors thank Téo Lemane for developing `kmindex` and for his responsiveness in addressing
 114 feature requests and issues raised during the development of `kmhelpers`. We acknowledge the
 115 GenOuest core facility (<https://www.genouest.org>) for providing the computing infrastructure.

The work was funded by the Inria Challenge “OmicFinder” (<https://project.inria.fr/omicfinder/>), and by the state funding managed by the French National Research Agency under the France 2030 program [ANR-22-PEAE-0005].

AI Usage Disclosure

AI assistance was used for constrained tasks (drafting, editing, code suggestions) under strict human review at every stage. AI provider used: Claude (Anthropic, 2025).

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